

EEE3094S 2026

Tutorial Test 1

Name: _____

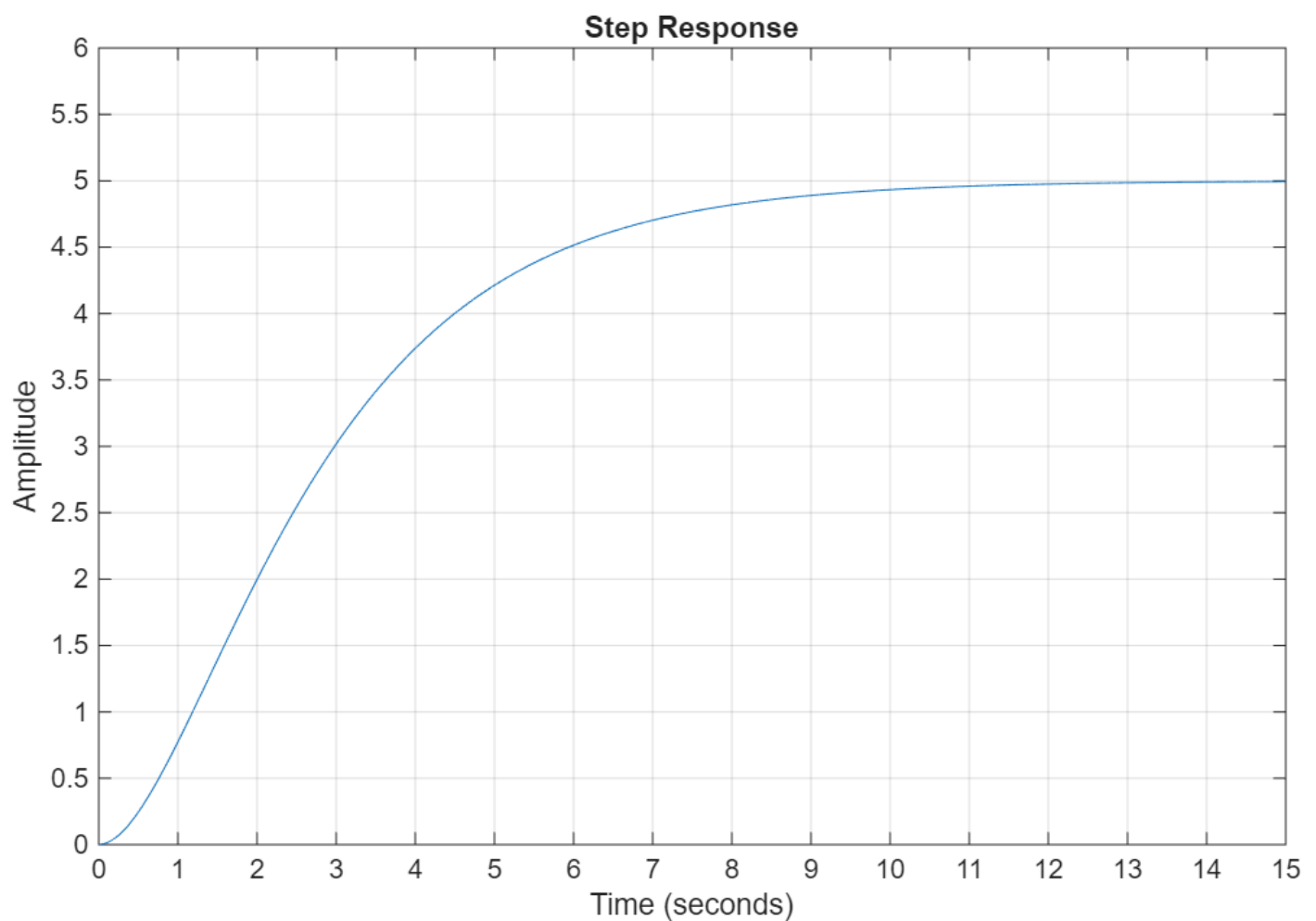
Student Number: _____

Instructions:

- The test is open book. You do not need to hand in your resources.
- You may do your diagrams in pencil but please ensure that your lines are clear.
- You have 20 minutes to complete the test + 5 minutes of reading time.

1. Time domain system analysis (11 points)

Analyse the following unit step response for an unknown order system and answer the questions following.



a) (1 point) What is the gain of the system?

b) (4 points) This system is either first or second order. Which one do you suspect, and why? Argue your assumption by measuring typical metrics for these systems (hint: τ values).

c) (2 points) Assuming no further information is available about the system, is it feasible to derive a transfer function by hand (i.e. excluding any fitting tools)? If not, what other information from the system would allow for modelling by hand?

d) (4 points) Irrespective of your answers to b) and c), suggest a first order model of the system, and sketch its unit step response on the plot below:

